



Short Communication

Novel linear plasmids carrying *vanA* cluster drives the spread of vancomycin resistance in *Enterococcus faecium* in India

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ABSTRACT

Objectives: The genetic basis for the spread of vancomycin resistance in *Enterococcus faecium* is largely unexplored in India. The present study aimed to investigate the plasmid diversity and variation of Tn1546 associated with *vanA* harbouring VREfm isolates.

Methods: A total of 122 VREfm isolates collected from blood cultures were included in this study. MLST analysis was performed on all isolates, and they were also screened for the presence of *vanA* and *vanB* genes. Whole genome sequencing was performed for a subset of fifteen VREfm isolates belonging to ST1643.

Results: All of the 122 VREfm isolates carried the *vanA* gene. Twenty-four different sequence types were seen; of these, ST1643, ST80 and ST17 were predominant. Whole genome sequencing was performed on 15 VREfm isolates belonging to ST1643. For eight isolates the *vanA* gene was found on pRUM-like circular plasmids, and for the remaining seven isolates, the *vanA* gene was found on the linear plasmids. Novel Tn1546 variants carrying *vanA* were found on both circular and linear plasmids. Interestingly, co-presence of *vanA* and *optrA* were seen in the backbone of three linear plasmids.

Conclusion: Multiple *vanA*-carrying plasmids and Tn1546-like elements were involved in the dissemination of vancomycin resistance in VREfm. The co-occurrence of Tn1546 carrying *vanA* and Tn554 family transposon carrying *optrA* on the backbone of plasmids is worrisome. The dissemination of such plasmids may pose treatment and infection control challenges.

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1. Introduction

The success and transition of *Enterococcus faecium* to a multidrug resistant nosocomial pathogen has been associated with the acquisition of various mobile genetic elements carrying antimicrobial resistance genes. Vancomycin resistant enterococcal (VRE) infections are difficult to treat owing to the limited treatment op-

tions, and they are associated with higher mortality rates [1]. Many vancomycin resistance gene clusters (*vanA*, *vanB*, *vanD*, *vanE*, *vanG*, *vanL*, *vanM* and *vanN*) have been described [2]. Among these, *vanA* and *vanB* are predominant, and they are often carried on the transposons Tn1546 and Tn1549, respectively [3]. The *vanA* gene cluster confers high-level resistance to both vancomycin and teicoplanin, whereas *vanB* results in resistance only to vancomycin. The *vanA*-carrying transposon Tn1546 is mostly found on conjugative plasmids such as Inc18-like, pRUM-like, pMG1-like and pHT-like, which are responsible for its widespread dissemination [4].

The majority of vancomycin-resistant *E. faecium* (VREfm) strains detected in hospitals belong to a single genetic lineage of clonal complex-17. Multiple sequence types (STs) are implicated in the global dissemination of VREfm strains [2]. In India, VREfm-

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associated bloodstream infection occurs at a rate of 42.2% of *E. faecium* bloodstream infection and has been documented with multi-clonal transmission belonging to ST80, ST17, ST117, ST132 and ST280 [5]. However, plasmids involved in the spread of vancomycin resistance are unexplored in India, and tracking of mobile genetic elements involved in the dissemination of VREfm genotypes has not yet been investigated. The present study aimed (i) to investigate the lineage of VREfm-causing bloodstream infections and (ii) to identify *vanA*-carrying plasmids using comparative genomic analysis of VREfm isolates.

2. Materials and methods

2.1. Bacterial strains

E. faecium isolates that are resistant to both vancomycin and teicoplanin with the minimum inhibitory concentration (MIC) of ≥ 32 µg/mL were included in this study. Nonduplicate isolates of VREfm (n=122) causing bloodstream infections collected between 2016 and 2020 from two hospitals, Christian Medical College, Vellore and All India Institute of Medical Sciences, Trauma Centre, New Delhi were included in this study. The isolates were identified as *E. faecium* using matrix-assisted laser desorption ionisation time-of-flight mass spectrometry (BioMérieux, Marcy-l'Étoile, France). The MIC of vancomycin, teicoplanin and linezolid were determined using the Clinical and Laboratory Standards Institute recommended broth microdilution method [6]. *Enterococcus faecalis* ATCC 29212 was used as the quality control strain for MIC testing.

2.2. Molecular characterisation

Genomic DNA of VREfm isolates was extracted from an overnight culture (14–16 h) grown on blood agar at 37°C, using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's protocol. The quality and concentration of the extracted DNA were assessed using NanoDrop One spectrophotometry (Thermo Fisher Scientific, MA, USA) and Qubit 3.0 Fluorometry (Life Technologies, CA, USA) and stored at –20°C.

All VREfm isolates were screened for the presence of *vanA* and *vanB* genes as defined earlier [7]. Multilocus sequence typing (MLST) was performed on all VREfm isolates (n=122) as described previously [8]. Sequenced data were analysed for STs using the *E. faecium* pubMLST database (<https://pubmlst.org/organisms/enterococcus-faecium>).

2.3. Genome sequencing, assembly and phylogenetic analysis

Whole genome sequencing was performed on a subset of 15 VREfm isolates. For short read sequencing, libraries were prepared using Illumina® Nextera® XT DNA Library Prep Kit (Illumina, San Diego, California, USA), according to the manufacturer's protocol. Sequencing was performed on a MiSeq platform using 300-cycle Reagent Kit v.2. For long read sequencing, Oxford Nanopore MinION sequencing device was used with FLO-MIN106 R9 MinION flow cells (Oxford Nanopore Technologies, Oxford, UK). A long-read DNA library was prepared using a SQK-LSK108 Ligation Sequencing Kit v.R9 along with an ONT EXP-NBD103 Native Barcode Expansion kit (Oxford Nanopore Technologies) as per manufacturer's protocol. The library was loaded onto the flow cells and was run for 48 h using the standard MinKNOW software. The Fast5 files generated from MinION sequencing were subjected to base calling with Guppy (v.4.4.2).

To obtain the complete circular genome, hybrid de novo assembly, quality checking and associated assembly statistics were carried out from both Illumina and MinION reads as described previously [9]. Unicycler hybrid assembly pipeline v.0.4.6 for prokaryotic

genomes with default settings was employed for hybrid de novo assembly using both Illumina and MinION reads. By default, Unicycler utilises SPAdes to assemble the short reads along with error correction and quality checks. The genomes were further polished with Pilon (v.1.22). The quality of assembly after each polishing step was assessed for completeness, correctness and contiguity metrics using CheckM (v.1.0.5) and Quast (v.4.5).

Sequence reads were mapped to the *E. faecium* reference genome Aus0004 (accession no. CP003351) using BWA mem algorithm [10]. Gubbins was used to remove regions of recombination [11]. Core genome single nucleotide polymorphisms (SNPs) were estimated using Snippy v.4.5.1 and filtered using FreeBayes [12,13]. The RAxML was used to generate a maximum-likelihood (ML) tree based on 931 variant sites in the core genome with 100 bootstraps and a midpoint root [14].

2.4. Genome annotation and downstream analysis

The genome was annotated using the Prokaryotic Genome Annotation Pipeline (PGAP) v.4.1 (https://www.ncbi.nlm.nih.gov/genome/annotation_prok/). Chromosomal mutations and resistance genes were identified using ResFinder 4.1 (<https://cge.cbs.dtu.dk/services/ResFinder/>). The plasmids were screened for the presence of highly conserved replication initiation (*rep*) gene using PlasmidFinder 2.1 (<https://cge.cbs.dtu.dk/services/PlasmidFinder/>). Tn1546 variants were identified by manual curation using Tn1546 prototype of *E. faecium* BM4147 (accession no. M97297) as the reference sequence. Genetic backbones of the linear plasmids were compared and depicted using *E. faecium* AA708 plasmid pELF1 (accession no. LC495616.1) as the reference genome using Easyfig (v.2.2.5).

2.5. Data availability

All the sequenced genomes were deposited in the Genbank under the accession numbers listed in the Supplementary Table S1.

3. Results

All VREfm isolates (n=122) carried *vanA* gene and showed vancomycin and teicoplanin MICs ranging from 32 to >512 µg/mL and 16 to 256 µg/mL, respectively. Among this VREfm collection, three isolates were also resistant to linezolid (MICs 16 to 32 µg/mL). A total of 24 distinct STs were identified, all except four belonging to clonal complex 17 (Supplementary Table S2). The most common were ST1643 (n=62), ST80 (n=28) and ST17 (n=13). ST1643 is the single locus variant of ST80 at the loci *pstS*. As strains of ST1643 were predominant, a subset of 15 VREfm (belonging to ST1643) isolates were studied in detail to understand the acquisition of vancomycin resistance encoding genes using genomic investigation.

3.1. Genome sequencing of ST1643 VREfm isolates

Hybrid assembly of sequencing reads from the Illumina and Oxford Nanopore MinION platforms yielded high-quality complete genomes of VREfm (n=15) that were further analysed for the presence and genetic background of the *vanA* determinant. The 15 sequenced VREfm isolates harboured between 5 and 12 plasmids (Supplementary Table S1). The different combinations of chromosomal mutations and their acquired resistance that conferred resistance to various classes of antibiotics are listed in the Supplementary Table S3. A phylogenetic tree for the 15 VREfm isolates belonging to ST1643 was constructed based on the 931 SNPs is shown in Supplementary Fig. S1, and the genomes were found to be genetically distinct.

3.2. Genetic backbone of *vanA*-carrying circular plasmids

In all of the 15 VREfm isolates, the *vanA* gene was found on plasmids. Eight of them were found to be closed circular plasmids beginning with primary replication initiation gene *rep17* in 5′–3′ direction (Supplementary Fig. S2) and a toxin-antitoxin (TA) system as a stability module. The combination of stability modules TA_{relE} with TA_{Fst}, TA_{Phd/YefM}, and TA_{Txe/YoeB} was found in five *vanA*-carrying plasmids (accession no. CP072581.1, CP072569.1, CP059825.1, CP071960.1, CP059767.1). Whereas in three plasmids (accession no. CP059818.1, CP059749.1 and CP059742.1), the stability modules were a combination of TA_{relE} with TA_{PemK/MazF}. All of these circular plasmids shared at least 99% similarity to that of previously reported *vanA*-carrying plasmids with a query coverage varying from 87% to 100% (Supplementary Table S4). Four of the *vanA*-carrying plasmids (accession no. CP072581.1, CP072569.1, CP059825.1, CP071960.1) also carried other resistance genes such as *aph*(3′)-III, *ant*(6)-Ia and *ermB* (Supplementary Table S3).

3.3. Genetic backbone of *vanA*-carrying linear plasmids

The remaining seven *vanA*-carrying plasmids showed no sequence identity for any of the *rep* genes described in the existing PlasmidFinder database. Also, these plasmids did not contain any relaxases genes that were typically present on the *vanA*-carrying plasmid. Plasmid sequence analysis revealed the presence of two CDS and a *repB* gene encoding replication initiation proteins in five (accession no. CP059785.1, CP059807.1, CP072588.1, CP072722.1, CP072733.1) of the seven plasmids (Supplementary Table S5). The backbone of these linear plasmids was highly similar to the linear reference plasmid pELF1 (accession no. LC495616.1) with an identity of 96.6% to 99.6% and a query coverage range of 74% to 85% (Supplementary Table S6). The remaining two *vanA*-carrying linear plasmids (accession no. CP059757.1 and CP072712.1) had a copy of genes involved in replication initiation (*dinB1*), as well as in DNA partitioning and transfer (*soj* or *relE*) (Supplementary Table S5). These two plasmids showed a low query coverage (22% and 62%, respectively) against the pELF1 plasmid (Supplementary Table S6).

In comparison to pELF1, none of the seven linear plasmids had a hairpin structure with a 5′-TATA-3′ loop at the left end. Further, m-fold analysis revealed the presence of palindromic sequences at the right end of all seven linear plasmids. The topology of these seven plasmids was determined to be linear based on the presence of the gene markers involved in plasmid replication (*repB*), DNA partitioning and transfer (*soj*, *ftsK*, *relE*) and the similarity of their backbone to pELF1 plasmid (Fig. 1 and Supplementary Table S5). Further, BLAST similarity analysis of these seven plasmids showed their sequence identity against several *E. faecium* plasmids, with a maximum score for all but one corresponding to linear plasmids (Supplementary Table S7). Interestingly, co-occurrence of vancomycin resistance conferring *vanA* associated with Tn1546 variant and linezolid resistance conferring *optrA* associated with Tn554 family transposon were found in the backbone of three linear plasmids (Fig. 2). Of them, Tn554 family transposons additionally carried the *ermA* gene in only two plasmids (accession no. CP059785.1 and CP072588.1) (Fig. 2).

3.4. Tn1546 variants in circular plasmids

Compared with typical Tn1546, the variants differed by partial or complete deletion of transposase genes as well as the position and type of IS elements located within the transposons. Furthermore, point mutations in both *vanRS* and *vanHAX* operon resulted in additional variation in Tn1546.

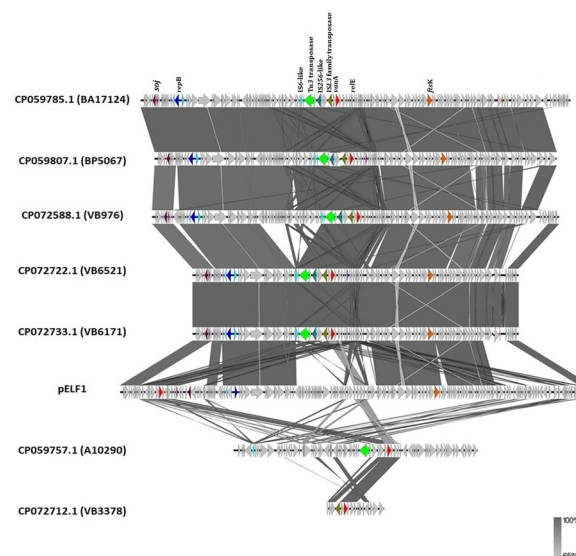


Fig. 1. Genetic backbone of *vanA*-carrying linear plasmids identified in ST1643. Regions shared between sequences are connected with vertical blocks. A minimum blast length of 500 bp and minimum identity of 80% was considered. The coding sequences are shown with arrows, and the colour reflects the genes' function: *vanA* genes are in red colour, putative genes involved in DNA replication are in dark blue (*repB*) and sky blue (replication initiation protein), genes involved in DNA partitioning and transfer are given in dark purple (*soj*) and orange (*ftsK*).

In the circular plasmids (*n*=8), the *vanA* gene was found on Tn1546-like element on a pRUM-like plasmid backbone. This novel Tn1546-like transposon lacked transposase and resolvase but did have an insertion sequence and transposase (IS6-like element and ISL3 family transposase) (Fig. 3a). IS6-like element was located at the upstream of *vanRS*, and ISL3 family transposase was inserted in the *vanS-vanH* intergenic region in reverse orientation. The *vanR* binding and interacting sites were uninterrupted by the insertion of IS6-like element within Tn1546. All of the VREfm isolates had the substitution V227A in the *vanA* gene, and in addition the isolate A11051 had E320V substitution in *vanH*.

3.5. Tn1546 variants in linear plasmids

Among the linear *vanA*-carrying plasmids (*n*=7), three distinct novel Tn1546 variants were identified (Fig. 3b). Tn3 family transposase and resolvase were found in all except one isolate (VB3378), which lacks Tn3 family transposase compared to the reference nucleotide sequence of the transposon Tn1546 (accession no. M97297). Deletion of 14 amino acids (at position: 975-KVVSLSNLRPLKLS-988) was noticed in the transposase of five plasmids (Supplementary Table S5). In five isolates, an insertion sequence IS256-like element was located at the upstream of *vanR*, and ISL3 family transposase was found in the intergenic region of *vanS-vanH* in reverse orientation. In an isolate A10290, IS1380 family transposase was found to be inserted at the upstream of *vanRS*, and there was no IS element at the *vanS-vanH* intergenic region. In VB3378 isolate, ISL3 family transposase was inserted at the *vanS-vanH* intergenic region. The binding and interacting sites of *vanR* remain intact with the insertion of IS256-like element and IS1380 family transposase. The substitution V227A was found in all *vanA* genes.

4. Discussion

VRE infections are on the rise in India, with the *vanA* cluster being the most common resistance mechanism [5]. To the best of our knowledge, this is the first study to report the whole genome

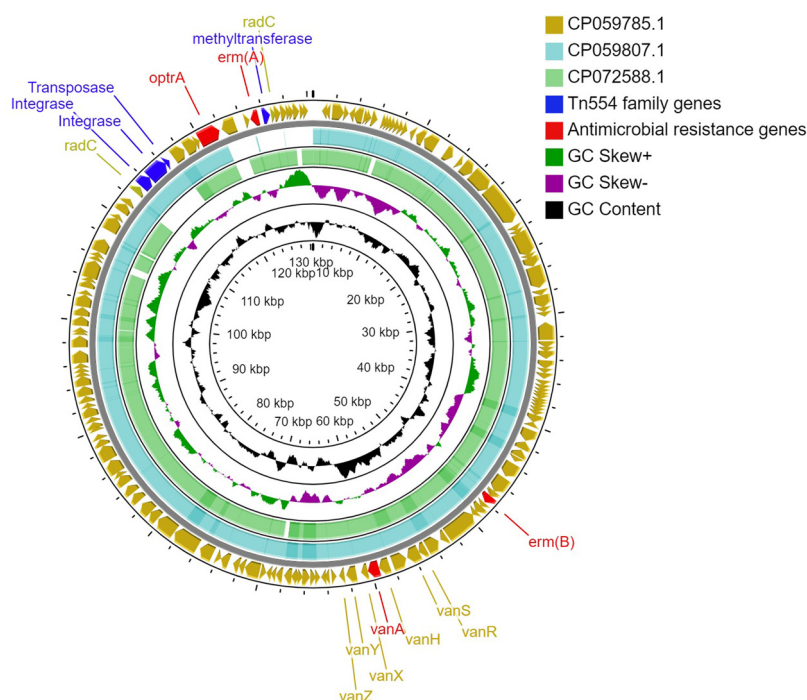


Fig. 2. BLAST analysis of three linear plasmids CP059785.1 (BA17124), CP059807.1 (BP5067), CP072588.1 (VB976) co-carrying vancomycin resistance expressing *vanHAX* operon associated with Tn1546 and linezolid resistance conferring *optrA* accompanying the Tn554 family transposon.

sequencing analysis of VREfm causing bloodstream infection from India. Here, we used complete genome sequencing to characterise the genetic background of vancomycin resistance in *E. faecium*.

In India, the multi-clonal spread of *vanA* type resistance in *E. faecium* has been documented with ST80 being predominant [5]. Similarly, in the present study, multiple STs were found. A substantial number of isolates belong to a VREfm lineage ST1643 has not been previously reported. This could be due to either the acquisition of conjugative plasmids or Tn1546 variants carrying *vanA* in multiple STs of *E. faecium* with different genetic background or multiple introductions of VREfm into hospitals. Several studies have reported Tn1546 variants on Inc18, pRUM-like, pMG1-like, pLG1 and pHT β -like plasmids [15]. Structural variations of Tn1546 linked with the presence of insertion sequences (IS1216, IS1251, ISEfa4, ISEfa5, ISEfm2, ISEf1, IS3-like, ISEfm1-like) at various locations have also been reported [16]. Notably, Tn1546 variants identified in this study were found to be novel and have not been previously described.

Many linear plasmids have been reported in *Streptomyces* spp., *Nocardia*, *Rhodococcus* and *Mycobacterium* [17–19]. In recent years, transferrable linear plasmids seem to have become increasingly important in the dissemination of Tn1546 among *E. faecium* isolates. In *E. faecium*, linear *vanA* plasmids such as pELF1, pELF2 and pIHVA have been reported from Japan [20–22]. These transferable plasmids are novel with a broad host range and completely different from that of *vanA*-carrying circular plasmids Inc18-, pRUM-, pMG1- and pHT β -like. In this study, the *vanA* gene cluster was found on both pRUM-like and linear plasmids, suggesting that it could lead to the selective expansion and hospital adaptation of the ST1643 lineage. In enterococci, linear plasmids have two different terminal structures with a hairpin-like loop interspaced by 5'-TATA-3' at the left end and palindromic sequences forming secondary loop structures at the right end [22]. In the present study, short-read coverage around the hairpin-like structure at the left end of the linear plasmids generated by Illumina Miseq was low,

making it difficult to locate the hairpin structure with a 5'-TATA-3' loop.

Hashimoto et al. reported the first enterococcal conjugative linear plasmid pELF1 that carried *vanA* and *vanM* resistance clusters [22]. Similarly, the *vanA* cluster was carried with a Tn1546-like element on the plasmids pELF2 and pIHVA, and the gene *vanZ* was interrupted by IS1216, resulting in the loss of teicoplanin resistance [20,21]. However, in the present study, the genes *vanR*, *vanS* and *vanH* located on Tn1546 variants remained undisturbed with the insertion of IS256-like element and ISL3 family transposase. In addition, Tn554 family transposon co-carrying the resistant genes *ermA* and *ant* (9)-Ia on the plasmids pELF2 and pIHVA have also been reported [20,21]. Interestingly, three of the linear plasmids found in this study had Tn554 family transposon carrying linezolid resistant gene *optrA* and intergated into the target site *radC* in its backbone. This is the first study that reports Tn554 family transposon carrying *optrA* on the linear plasmid. In enterococcal linear plasmids, *repB* of the Rep_2 superfamily is involved in DNA replication and the gene *ftsK* is involved in plasmid transfer [20–22]. Of the seven linear plasmids identified in this study, five had a combination of putative genes (*repB*, two CDS of replication initiation protein, *dinB1*, *ftsK*, *soj* and *relE*), indicating that their replication and transfer mechanisms were very similar to pELF1, pELF2 and pIHVA.

The multi-clonal spread of the VREfm genotype was observed in this study collection. Genome analysis revealed the presence of the *vanA* cluster on both pRUM-like and linear plasmids in ST1643, which may contribute to its clonal expansion. Novel variants of Tn1546 associated with the presence of insertion elements IS256-like element and ISL3 family transposase were found in both pRUM-like and linear plasmids. The co-localisation of vancomycin resistance conferring *vanA* and linezolid resistance conferring *optrA* on the same linear conjugative plasmid is alarming. Therefore, the spread of such plasmids could severely limit the therapeutic options for VRE infections.

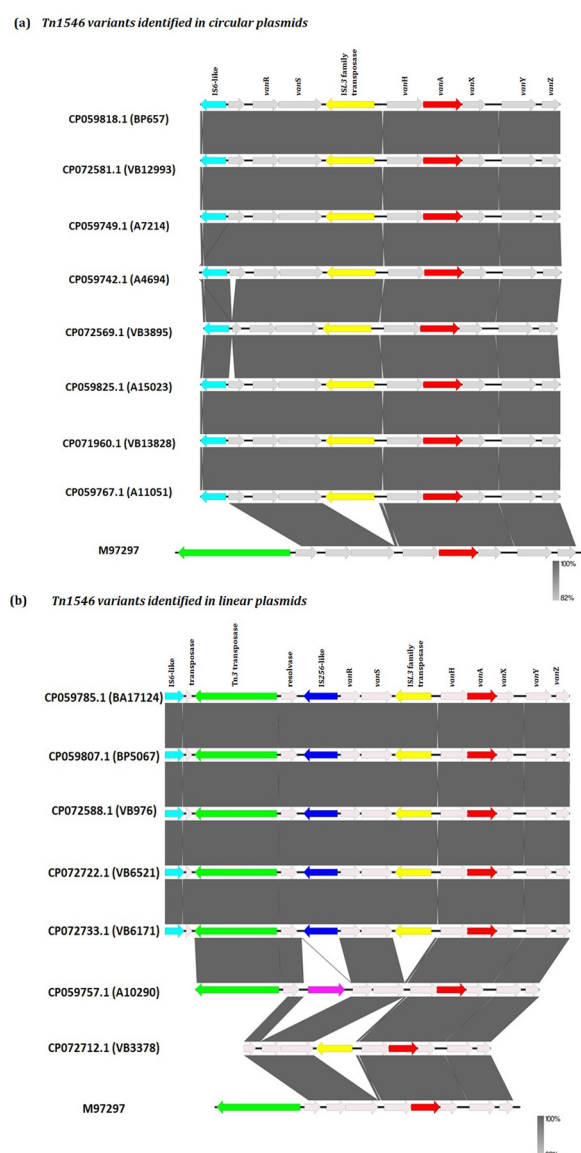


Fig. 3. Novel Tn1546 transposon variants identified in the circular plasmids (a) and linear plasmids (b) of VREfm belonging to ST1643. In Fig. 3b, the pink arrow (A10290) denotes the presence of IS1380 family transposase.

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Competing interests

None declared.

Ethical approval

None required.

Supplementary materials

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